Assignment Title: Guided Inquiry - Understanding Standard Volumetric Flowrates

Instructions:
In this assignment, you will use an online simulator to explore the relationship between different types of gas flowrates. Your goal is to understand what "standard volumetric flowrate" means and why it is a critical concept in science and engineering.

1. Open the Gas Flow Simulator in a new tab: [Insert Simulator Link Here]
2. Follow the steps outlined in each part below.
3. For each question that begins with **"Your Response:"**, type your answer directly into the corresponding text box for this assignment.

Part 1: First Observations & Predictions

First, let's get familiar with the simulator. You can adjust five variables: **mass flow**, **actual volumetric flow**, **standard volumetric flow**, **temperature**, and **pressure**.
Instructions:
Set the **mass flowrate** to 10 kg/s. Use the default values for temperature and pressure.
Question 1.1 (Your Response):
What are the initial values for the **actual volumetric flowrate** and the **standard volumetric flowrate**? Are they the same or different?
Question 1.2 (Prediction - Your Response):
In your own words, what do you predict is the difference between the *actual* volumetric flowrate and the *standard* volumetric flowrate? What does the word "standard" imply to you in this context?

Part 2: The Effect of Temperature & Pressure

Now, we will investigate how the physical conditions of a gas affect how we measure its flow.
Instructions:
Set a constant **mass flowrate** of 8 kg/s. This value represents a fixed amount of gas molecules moving through the system each second. Do not change the mass flowrate for the rest of Part 2. Note the starting values for all variables.
Investigation 2A: Temperature
Question 2.1 (Prediction - Your Response):
If you heat the gas (increase temperature) while keeping the mass flow and pressure constant, what do you predict will happen to the **actual volumetric flowrate**? Will it increase, decrease, or stay the same? What do you predict will happen to the **standard volumetric flowrate**?
Instructions:
Now, test your prediction. Slowly increase the **temperature** while watching the bar chart and the numbers.
Question 2.2 (Explain - Your Response):
After running the experiment, explain *why* the actual volumetric flowrate changed as it did. Thinking about gas particle behavior, why did the mass flowrate and standard volumetric

flowrate remain constant?

Investigation 2B: Pressure

Instructions:

Reset the temperature to its original value. Your mass flowrate should still be 8 kg/s.

Question 2.3 (Prediction - Your Response):

If you compress the gas (increase pressure) while keeping the mass flow and temperature constant, what do you predict will happen to the **actual volumetric flowrate**?

Instructions:

Now, test your prediction. Slowly increase the **pressure**.

Question 2.4 (Explain - Your Response):

After running the experiment, explain *why* compressing the gas changed its actual volumetric flowrate. Why is it useful to have a measurement like standard volumetric flowrate that *doesn't* change with pressure?

Part 3: Finding the "Standard"

What makes the standard volumetric flowrate "standard"? Let's find out.

Instructions:

Your goal is to adjust the **temperature** and **pressure** until the **actual volumetric flowrate** is exactly equal to the **standard volumetric flowrate**. The bars for these two values on the chart should be the same height. (You can set any mass flowrate to start).

Question 3.1 (Analysis - Your Response):

At what specific temperature and pressure did you find that the actual volumetric flowrate equals the standard volumetric flowrate?

Question 3.2 (Analysis - Your Response):

These values are known as "Standard Conditions" (or STP/NTP). Why do you think scientists and engineers need a universal baseline like this for comparing gas measurements?

Part 4: Putting It All Together

Based on your observations, let's summarize the key concepts.

Question 4.1 (Your Response):

Based on your observations, define the following terms in your own words:

a) Actual Volumetric Flowrate:

b) Standard Volumetric Flowrate:

Question 4.2 (Application - Your Response):

Consider this scenario: An engineer needs to ensure two factories, one in cold Minnesota (-10°C) and one in hot Texas (35°C), receive the exact same amount of natural gas fuel. Which measurement—**actual** or **standard** volumetric flowrate—should the engineer use to compare the gas deliveries? Explain your reasoning.

Question 4.3 (Conclusion - Your Response):

In your own words, summarize the primary reason why standard volumetric flowrate is an essential concept in science and engineering for measuring and comparing gas quantities.

End of Assignment

Submission: Once you have answered all questions, please submit your assignment through

Blackboard.